

Course: B. Tech.

Branch : Civil Engineering

Semester : IV

Subject Code & Name: BTCVC402 / BTCIC402 Environmental Engineering

Max Marks: 60

Date: 14/06/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.		12
A) Enlist various methods of population forecasting. Explain any one in detail.	CO1	6
B) Explain factors affecting rate of water demand.	CO1	6
C) Write a note on characteristics of water.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Explain different type of valves in detail with neat sketches.	CO2	6
B) Write a note on operational filter troubles.	CO2	6
C) Explain break point of chlorination with figure.	CO2	6
Q. 3 Solve Any Two of the following.		12
A) Explain systems of water supply with neat sketches.	CO3	6
B) Write a note on the layout system for water distribution.	CO3	6
C) Explain hydraulic analysis of distribution system.	CO3	6
Q.4 Solve Any Two of the following.		12
A) Explain combined system of sewerage.	CO4	6
B) Write a detailed note on Composting and factors affecting composting.	CO4	6
C) Explain concept of hazardous waste management.	CO4	6
Q. 5 Solve Any Two of the following.		12
A) Write a note on Electrostatic Precipitator.	Remember	6
B) What is Air Pollution? What are the major sources of Air Pollution?	Remember	6
C) Explain with suitable diagrams: Plume Behaviour.	Remember	6

*** End ***